INCH- POUND
MIL-C-71030 (AR)
24 June 1992

MILITARY SPECIFICATION

CARTRIDGE, IGNITION, M981
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements and quality assurance provisions for loading, assembling and packing of ignition cartridges designated as Cartridge, Ignition, M981.

2. APPLICABLE DOCUMENT

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (See 6.2).

SPECIFICATIONS

MILITARY

MIL-P-223 - Powder, Black
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-P-71032 - Primer, Percussion, M132 Loading, Assembling and Packing

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 1315
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

MIL-STD-414 - Sampling Procedures and Tables for inspections by variables for percent defective

(Unless otherwise indicated copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those cited in this solicitation.

DRAWINGS (See 6.6)

U.S. ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

12577526 - Cartridge, Ignition, M981 Assembly
12577527 - Primer Housing Assembly
12577533 - Percussion Primer Assembly
12577534 - Percussion Primer Housing Assembly
12952125 - Packing and Marking for Box, Fiberboard Packing for Cartridge, Ignition, M981

INSPECTION EQUIPMENT DRAWINGS

9278622 - Envelope Drawing for Leak Testing of Ignition Cartridges

(Copies of drawings and other publications required by manufacturers in connection with specified acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

PUBLICATIONS

CODE OF FEDERAL REGULATIONS

49 CFR Interstate Commerce Commission Rules and Regulation for the Transporation of Explosives and other Dangerous Articles

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402. Orders for the above publication should be cite 49 CFR (latest revision)).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specified exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Assembly. The ignition cartridge assembly and components shall comply with all requirements specified on Drawing (dwg) 12577526, associated drawings, and with all requirements specified in applicable specifications, and standards.

3.3 Moisture content of black powder. The moisture content of the black powder used in loading the ignition cartridge at the time of loading, shall not exceed 0.70 percent. Testing shall be as specified in 4.5.1.

3.4 Sensitivity.

3.4.1 Nonfunctioning. The primer shall not function under an applied energy level of 4.5 inch ounces minimum (min) when tested as specified in 4.5.2.1.

3.4.2 Functioning. The primer shall function under an applied energy value of 82 inch ounces maximum (max.) when tested as specified in 4.5.2.2.

3.4.3 Safety drop test. The primer shall not function as a result of the 6.5 foot cartridge drop specified in 4.5.2.3.

3.5 Leak test. The loaded cartridges shall satisfactorily pass the leak test specified in 4.5.3.

3.6 Velocity, functioning and security (proving ground test). The ignition cartridges shall comply with the following requirements (see 4.5.5).

3.6.1. Misfires. The ignition cartridges shall function on the first attempt, without any audible hangfire.

3.6.2. Metal part security. There shall be no evidence of rupture or metal parts separation attributable to ignition cartridge Functioning.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

3.6.3. Velocity and functioning (proving ground). The cartridge assemblies shall function and shall impart a corrected muzzle velocity of 101 ± 3 mps. In addition, the velocity standard deviation of the lot shall be not greater than 1.6 mps.

3.7. Propellant charge uniformity. The XM44 propellant lots shall achieve a corrected mean velocity of 101 mps (see 4.5.6). This test shall be required when a lapse of two years or more occurs between propellant acceptance and LAP initiation.

3.8 Static pressure. The average internal ignition cartridge pressure plus four standard deviations shall not exceed 14,200 psi (see 4.5.7).

3.9 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with technical provisions herein (see 4.3 and 6.2 and 6.3).

3.10 Workmanship. All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government, The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. The requirements of MIL-A- 48078 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2. Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection (See 4.3)

b. Quality conformance inspection (See 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer in accordance with the provisions of 4.3.2. The first article sample shall consist of the items in the quantities indicated in Table I.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71030

PARAGRAPH	TITLE Cartridge, Ignition, M981 Assembly	SHEET 1 OF 1	DRAWING NUMBER See below
			NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Percussion primer subassembly			
	(dwg. 12577534)			
	Examination for defects	20	3.2	4.4.2.1
	Percussion primer assembly			
	(dwg. 12577533)			
	Examination for defects	20	3.2	4.4.2.2
	Primer housing assembly			
	(dwg. 12577527)			
	Examination for defects	20	3.2	4.4.2.3
	Cartridge, Ignition, M981 Assembly			
	(dwg. 12577526)			
	Sensitivity safety drop test	25	3.4.3	4.5.2.3
	Leak test	50	3.5	4.5.3
	Velocity, functioning and security	50	3.6	4.5.5
	Static pressure	50	3.8	4.5.7

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:34:17
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot of ignition cartridges shall contain:

a. Metal parts from one lot interfix from one manufacturer.

b. Black powder from not more than one lot from one manufacturer.

c. Percussion elements from not more than one lot interfix from one manufacturer.

d. M44 propellant from not more than one lot from one manufacturer.

e. O-rings from not more than one lot from one manufacturer.

4.4.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. Where specified herein, attributes sampling inspection shall be conducted in accordance with Table II below, using the inspection levels stated in the Classification of Characteristics paragraphs:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

TABLE II. Attributes sampling inspection

LOT SIZE	INSPECTION LEVELS I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	5	3
91 to 150	*	125	32	13	13	5
151 to 280	*	125	32	32	20	8
281 to 500	*	125	32	32	20	8
501 to 1200	*	125	80	50	20	13
1201 to 3200	1250	125	80	50	32	13
3201 to 10000	1250	125	125	50	32	13
10001 to 35000	1250	315	125	80	50	13
35001 to 150000	1250	315	125	80	50	13

Numbers under inspection levels indicate sample size; asterisks indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative quality conformance. Unless otherwise specified herein or provided for in the contract, alternate quality conformance procedures, methods or equipment such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternate procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.10). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provisions herein. In case of dispute as to whether the contractor's proposed alternative(s) provide equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

QUALTY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Percussion Primer Housing Assembly		12577534
			NEXT HIGHER ASSEMBLY 12577533

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Sensitivity nonfunction	1/	3.4.1	Test/4.5.2
2	Sensitivity functioning	1/	3.4.2	Test/4.5.2
Major				
101	Percussion primer missing	Level III	3.2	Visual
102	Percussion primer inverted	Level III	3.2	Visual
103	Lacquer seal missing or inadequate	Level III	3.2	Visual
104	Lacquer on primer face	Level III	3.2	Visual
105	Crimp missing or incomplete	Level III	3.2	Visual
Minor				
201	Evidnece of poor workmanship	Level V	3.10	Visual

NOTES: 1/ Sample per 4.4.3.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71030 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Percussion Primer Assembly	1 OF 1	12577533
			NEXT HIGHER ASSEMBLY: 12577527

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Weight of black powder	Level III	3.2	Gage
102	Moisture content black powder	1/	3.3	Gage
103	Adhesive missing,inadequate or excessive	Level III	3.2	Visual
Minor	None defined			

NOTES:

1/ Sample shall be selected from each four (4 hour production interval.

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71030 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Primer Housing Assembly	1 OF 1	12577527
			NEXT HIGHER ASSEMBLY: 12577526

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special A	Firing plug protrudes above head surface (internal)	100%	3.2	Gage
Major				
101	O-ring missing from firing plug	Level III	3.2	Visual
102	O-ring missing from head/sleeve	Level III	3.2	Visual
103	Thread sealant missing, inadequate or excessive	Level III	3.2	Visual
104	Sleeve assembly torque improper	Level III	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Cartridge, Ignition, M981 Assembly	1 OF 1	12577526
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Sensitivity safety drop test	25	3.4.3	4.4.3.2.3/4.5.2.3
Special				
A	Weight of propellant (before loading)	100%		Gage
B	Weidht of propellant (after loading)	1/	3.2	Gage
Major				
101	RTV sealant missing, inadequate, or excessive	Level III	3.2	Visual
102	Length, Max.	Level III	3.2	Gage
103	Crimp missing or incomplete	Level III	3.2	Visual
104	Leak test	100%	3.5	Gage
105	Velocity, functioning, and security (proving ground)	2/	3.6	
106	Marking missing or unidentifiable	Level III	3.2	Visual
107	True position of cup to thread	Level III	3.2	Gage
108	Environmentral conditions inadequate	3/	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:
1/ See 4.4.3.1.2, Same sample used for both inspections.
2/ See 4.4.3.3.
3/ Temperature and humidity shall be recorded continuously, and checked at the beginning and end of each shift. If conditions are not proper at either time, the quantity of product represented by the conditioning shall be rejected for further processing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

4.4.3 Testing.

4.4.3.1 Weight of propellant.

4.4.3.1.1 Before loading. (See dwg. 12577526.) The propellant charge shall be weighed. Weight limits for weighing should be established within the applicable requirements, depending upon the accuracy and precision of the weighing system to assure compliance with the charge weight requirements. Any charge of bulk propellant failing to meet the applicable requirements shall be classed defective and removed from the lot.

4.4.3.1.2 After loading. Weight 0.5 grams or less above assessed maximum weight - Special Defect: Weight more than 0.5 grams above assessed maximum weight - Special Defect: Weight 0.5 grams or less below assessed minimum weight - Special Defect: Weight more than 0.5 grams below assessed minimum weight - Special Defect. 315 sample cartridge assemblies shall be randomly selected from each lot for this test. If four or more samples exceed the assessed max weight by 0.5 or less grams or if any sample exceeds the assessed max weight by more than 0.5 grams, the lot shall be rejected. If four or more samples are under the assessed minimum weight by 0.5 grams or less or if any sample is under the assessed minimum weight by more than 0.5 grams, the lot shall be rejected. In addition, if the total number of defective samples are five or more, the lot shall be rejected. Testing shall be as specified in 4.5.4.

4.4.3.1.3 Automated method. The contractor may elect to perform this operation with automated equipment and mandatory process controls. In such cases, process control plans in accordance with Data Item Description DI-P-1604 must be submitted to the Technical Agency for approval of the automated operation.

4.4.3.2 Sensitivity

4.4.3.2.1 Nonfunctioning.

a. Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, 200 primer housing assemblies shall be randomly selected from each lot for this test. If one or more assemblies fail to comply with the specified requirement, the lot shall be rejected. The test shall be performed as specified in 4.5.2.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

b. After three consecutive lots have met the criteria of 4.4.3.2.1, the sampling plan for this test shall be in accordance with Level III. If a lot is rejected, revert to sampling plan per 4.4.3.2.1. The test shall be performed as specified in 4.5.2.1.

4.4.3.2.2 Functioning.

a. Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, 200 primer housing assemblies shall be randomly selected from each lot for this test. If 4 or more primers fail to comply with the specified requirement, the lot shall be rejected. If 1 but less than 4 defectives are found a second sample of 200 primer housing assemblies shall be tested. If the combined number of defectives in both the first and second sample is 4 or more, the lot shall be rejected. Any assemblies in the randomly selected sample missing a component or with an incorrectly assembled component which will result in an end item misfire shall be considered defective and shall be included in the rejection criteria. The test shall be performed as specified in 4.5.2.2.

b. After three consecutive lots have met the criteria of 4.4.3.2.2, the sampling plan for this test shall be per Level III and shall be randomly selected from each lot for this test. If 1 or more primers fail to comply with the specified requirement, the lot shall be rejected. Any assemblies in the randomly selected sample missing a component or with an incorrectly assembled component which will result in an end item misfire shall be considered defective and shall be included in the rejection criteria. If a lot is rejected, revert to sampling plan per 4.4.3.2.2. The test shall be performed as specified in 4.5.2.2.

4.4.3.2.3 Safety drop test. Twenty-five ignition cartridge assemblies consisting of the primer housing assembly and ignition cartridge cup (assembled without the M44 propellant sticks in accordance with dwg 12577526) shall be selected for this test. If one or more of the test samples function when tested as specified in 4.5.2.3, the primer lot shall be rejected. This test shall be performed only upon initial use of each primer lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

4.4.3.3 Velocity, functioning and security

a. Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, the sampling plan and procedure for this test shall be in accordance with MIL-STD-414, Section B, Table B-3, Code letter M and a maximum percent defective of 2.49. In addition, if the velocity standard deviation of the sample exceeds the lot requirement, the lot shall be referred. If any round fails to function for reasons attributable to ignition cartridge failure or if a metal parts separation occurs, the lot shall be rejected. This test shall be performed as specified in 4.5.5.

b. After three consecutive lots have met the criteria of 4.4.3.3.1.A the sampling plan for this test shall be in accordance with MIL-STD- 414, Section B, Table B-3, Code Letter K, using a maximum percent defective of 2.68. In addition, if the velocity standard deviation of the sample exceeds the applicable lot requirement, the lot shall be referred. If any round fails to function for reasons attributable to ignition cartridge failure or if a metal parts separation occurs, the lot shall be rejected. If a lot is rejected revert to sampling plan per 4.4.3.3.1.A. This test shall be performed as specified in 4.5.5.

4.4.3.4 Check test for deterioration of primer, percussion, M132. If the total elapsed time between original acceptance of the M132 primer lot and the assembly of that lot into the ignition cartridge exceeds two years, or if the M132 primers have been subjected to adverse conditions for any period of time since previous test, the M132 primer lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in MIL-P-71032 immediately before the M132 primer lot is assembled into the ignition cartridges.

4.4.3.5 Static pressure. The M44 propellant lot shall be tested as specified in 4.5.7. Twenty-five ignition cartridges shall be loaded at the assessed charge weight with M44 propellant randomly selected from the propellant lot. If the average ignition cartridge pressure exceeds the specified requirement, the representative lot of M44 propellant shall be withheld and referred to the engineering agency.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Inspection Method Reference" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.5 herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

4.5 Methods of inspection.

4.5.1 Moisture content of black powder. The moisture content of black powder shall be determined in accordance with the procedure specified in MIL-P-223. The samples shall be randomly selected from each four hours production. Any failure to comply with the applicable requirement shall be cause for rejection of the four hours production represented by the sample. If loading has begun, ignition cartridges loaded with the nonconforming black powder shall be rejected.

4.5.2 Sensitivity.

4.5.2.1 Nonfunctioning. The primer housing assembly shall be assembled to a suitable fixture. The firing pin of the test fixture shall be adjusted so that impact will be on the center of the firing plug of the primer. A steel ball shall be dropped onto the firing pin producing 4.5 in oz of energy (min). The height of drop shall be measured as the height from the top of the firing pin to the bottom of the weight. Any primer housing assembly which functions shall be classed defective. Samples used in this test shall not be returned to the lot.

4.5.2.2 Functioning. The primer housing assembly (12577534) shall be assembled to an appropriate test fixture. The firing pin of the test fixture shall be adjusted so that impact will be 0.05 inch off center of the firing plug of the primer. A steel ball shall be dropped on to the firing pin producing 82.0 in oz of energy (max). The height of drop shall be measured as the height from the top of the firing pin to the bottom of the weight. Any primer housing assembly that fails to function shall be classed defective.

4.5.2.3 Safety drop test. The test assemblies shall be assembled into inert-loaded, standard 120mm cartridges. Each cartridge shall be dropped twice from a height of 6.5 feet on an anvil top surface. For the first drop, the cartridge shall be in the base (fin) down orientation. For the second drop, the cartridge shall be in the nose (fuze) down orientation. The drop height shall be measured from the lowest point of the cartridge orientation to the top of the anvil surface. Any test assembly which functions as a result of the drop shall be classed defective. All questionable test assemblies shall be downloaded to verify primer functioning or nonfunctioning. All components used for this test shall not be returned to their respective lots.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

4.5.3 Leak test. The ignition cartridges shall be tested 100% in accordance with drawing 9278622, after a two (2) hour minimum cure time. Any ignition cartridge which fails to comply with applicable requirement shall be classed defective and removed from the lot.

4.5.4 Weight of propellant. 315 random samples of the ignition cartridge will be taken prior to crimping operation. All propellant removed and placed on the pan of a precision analytical balance. The weight of propellant from each sample shall be determined to the nearest 0.05 gram.

4.5.5 Velocity, functioning and security (proving ground test). The test shall be performed at a Government proving ground in accordance with the applicable USA TECOM Acceptance Test Procedure.

a. The sample ignition cartridges shall be assembled with a standard finned projectile (30.1 ± 0.066 pounds (lbs)) for which they were manufactured. The complete assembly, less propelling charges, shall be fired from a 120mm ballistic tube at an elevation of forty-five degrees.

b. Reference rounds shall be fired without propelling charges for the purpose of velocity correction. At least one reference round should be fired for every three test rounds.

c. Individual velocities shall be recorded on the firing record for both reference and test rounds. Velocity standard deviations and arithmetic averages shall be calculated and recorded. Temperature of the day and time of firing shall also be recorded. In the event of a misfire, the misfired ignition cartridge shall not be disassembled by the proving ground but shall be forwarded to the responsible technical activity. Individual chamber pressures shall be obtained for each round firing.

4.5.5.1 Refiring. If for any reason the proving ground considers that conditions have detrimentally affected the test results, additional cartridges as required shall be tested.

4.5.5.2 Refferal Criteria. Any test load fail velocity standard deviation or average pressure requirements at 20 degrees C or whenever the Proving Ground deems necessary.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

4.5.6 Propellant charge uniformity. (To be conducted at a U.S. Government Proving Ground). Ten test ignition cartridges shall be loaded at charge weight with a tolerance of plus or minus 0.5 grams. The ignition cartridges shall be assembled to 120mm production rounds. Rounds shall be inert loaded to achieve a total weight of 30.1 pounds plus or minus 0.066 pounds. The rounds shall be conditioned at 70 degrees F (+2.5 degrees F) for 24 hours prior to firing and shall be maintained at this temperature until fired. The rounds shall be fired from a 120mm ballistic tube at an elevation of forty-five degrees. Reference rounds shall be fired alternately for velocity correction. The mean corrected velocities of the group shall be compared to the established charge weight velocity. If the corrected velocity of the test rounds exceeds three percent of the expected muzzle velocity, use of the propellant lot shall be referred. Test results shall be forwarded to the technical agency for evaluation (see 6.5).

4.5.7 Static pressure test. The ignition cartridges shall be threaded into the XM31 Fin and firmly seated. A 5.5mm hole shall be drilled in the ignition cartridge tube through one flash hole in the sixth row of flash holes in the fin boom, counting from the ignition cartridge end. A pressure transducer, mounted to a collar (see figure 1), shall be attached to the fin boom positioning the collar so that the transducer will be directly over the flash hole drilled in the ignition cartridge tube. The collar shall be positioned carefully to prevent restriction of venting from other flash holes in the boom. The assembly shall be set up as shown in figure 2 and the transducer connected to a recorder. The ignition cartridge shall then be functioned and pressure/time tracer recorded.

5. PACKAGING

5.1 Packaging.

5.1.1 Level A. Not applicable

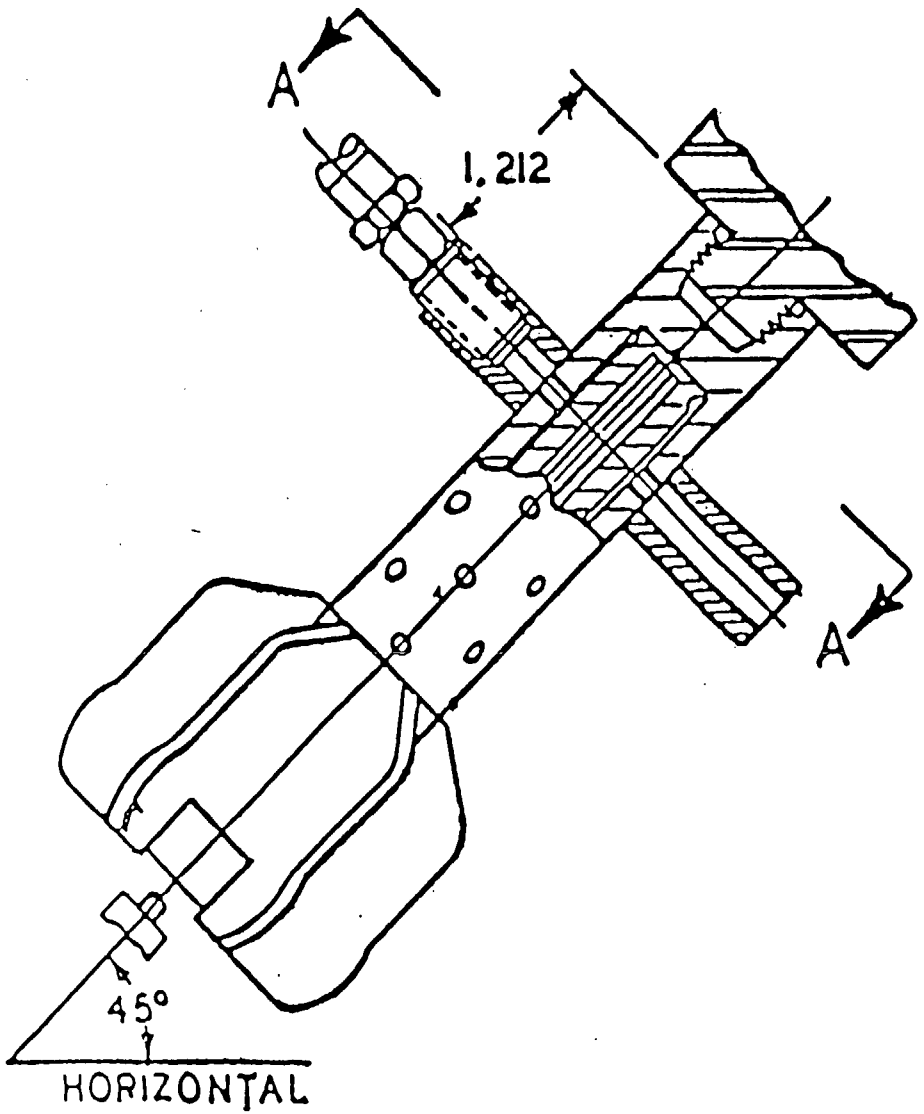
5.1.2 Level B. Exterior packing shall be as specified in dwg. 12952125.

5.1.3 Level C. Same as Level B.

5.2 Marking.

5.2.1 Level A. Not applicable

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.



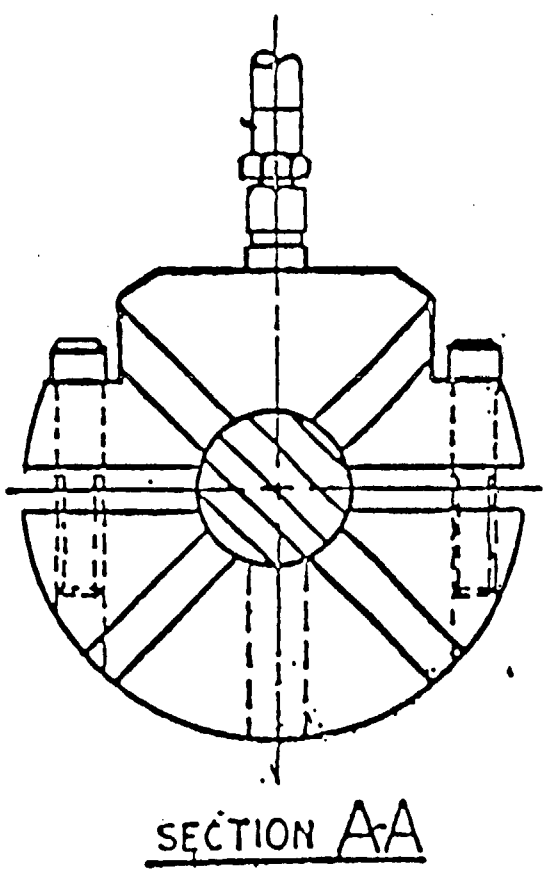


Figure 1. Collar

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

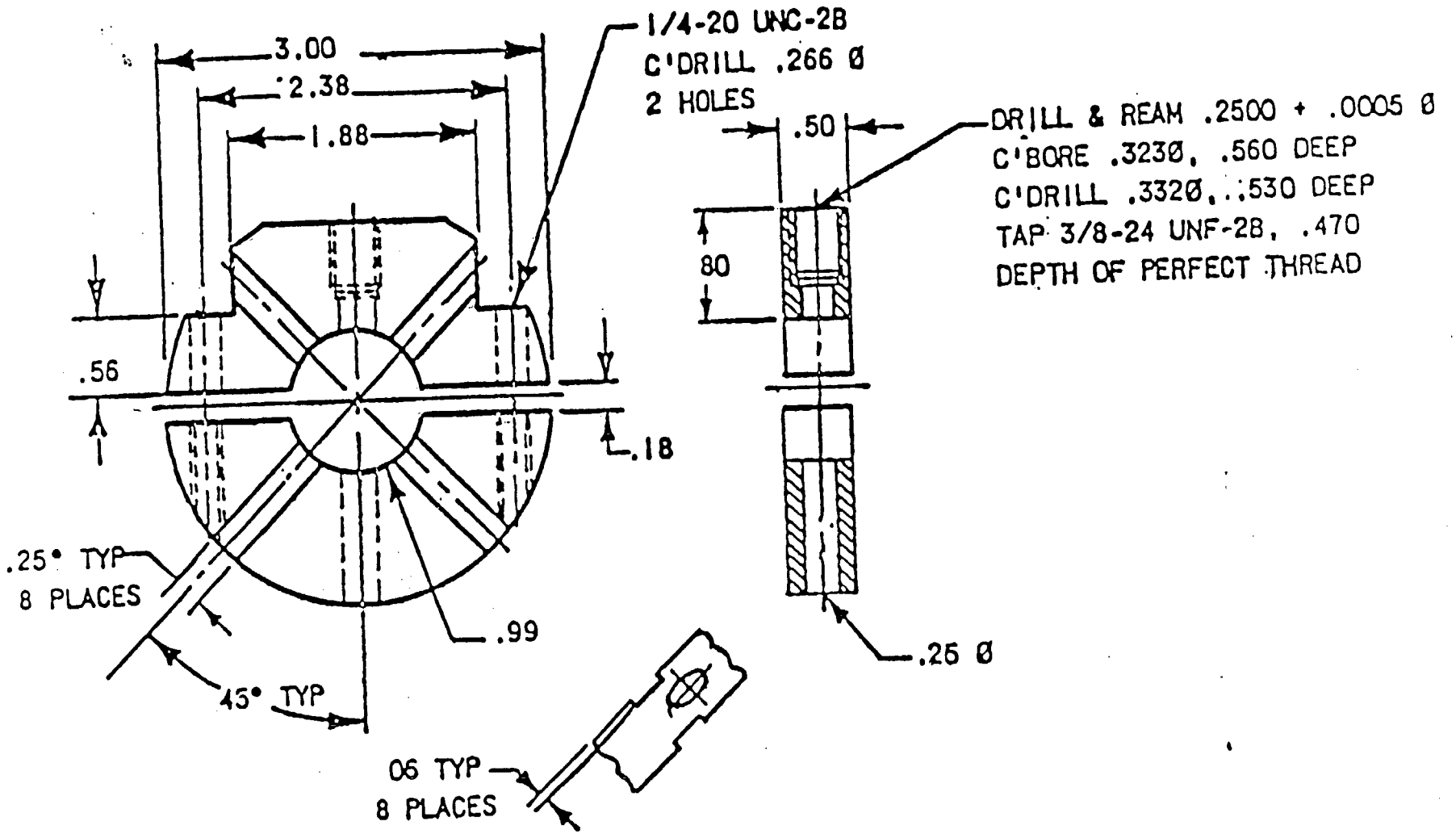


Figure 2. Static Ignition Cartridge Test Setup

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

5.2.2 Level B. Marking shall be as specified in 49 CFR and drawing 12925125.

5.2.3 Level C. Same as Level B.

6. NOTES

(This section contains information of general or explanatory nature may be helpful, but not mandatory.)

6.1 Intended use. The components covered by this specification are intended for assembly to M929/M930/M933/M934 cartridges for use with the 120mm mortar.

6.2 Acquisition requirements.

a. See MIL-A-48078.

b. Issues of DODISS to be cited in the solicitation and if required, the specific issue of the individual document referenced (see 2.1).

c. Provisions for submission of first article sample.

6.3 First Article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such product, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Consideration of Data Requirements. The following data requirements should be considered when this specification is applied to the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

Reference Paragraph	DID Number	DID Title
4.4.3.1.3	DI-P-1604	Process Control Plan

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited in the DD Form 1423.

(Copies of data item descriptions required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

6.5 Submission of inspection equipment designs for approval. See paragraph 6.4.3 of MIL-A-48078 (AR). Submit equipment designs, as required, to Commander, U.S. Army Armaments, Munitions and Chemical Command (ARDEC) Attn: SMCAR-QAT-I(D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the contract data requirements list DD firn 1423 in the contract.

6.6 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as ARRADCOM, Edgewood Arsenal, Frankfort Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.7 Submission of test data. One copy of proving ground ballistic acceptance test data (i.e., firing record) shall be forwarded to each of the following:

COMMANDER
U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT
AND ENGINEERING CENTER
ATTN: SMCAR-QAT-B (D)
SMCAR-FSA-IM
PICATINNY ARSENAL, NJ 07806-5000

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

6.8 Check test for possible deterioration of cartridge, ignition. The following test is designed to prevent the loading of major ammunition items with ignition cartridges which have deteriorated. The test shall be performed on each lot of ignition cartridges which have been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the ignition cartridge is assembled does not require that the check test be performed, the contracting officer shall insert a provision for the performance of the check test in the contract. The contracting officer shall arrange to have the contractor be reimbursed for any expense incurred in the performance of the check test for possible deterioration.

6.9 Velocity, functioning and security. Beginning with the first with the first lot produced and continuing until three consecutive lots have compiled with the applicable requirements specified. A sample from each lot shall be selected in accordance with MIL-STD-414, Section b, Table B-3, Code Letter M and a maximum percent defective of 2.49. After three consecutive losts have met the criteria of 4.4.3.3.1.a, the sampling plan for this test shall be in accordance with MIL-STD-414, Section B, Table B-3, Code letter K, using a maximum percent defective of 2.68. In addition, if the velocity standard deviation of the sample exceeds the lot requirement, the lot shall be rejected. If any round fails to function for reasons attributable to ignition cartridge failure, or if a metal parts separation occurs, the lot shall be referred. This test shall be performed as specified in 4.5.5.

6.10 Submission of alternative quality conformance provisions. Proposed alternative inspection provision should be submitted by the contractor to the procuring contracting officer for evaluation and approval by the technical activity responsible for preparation of this specification.

6.11 Subject term (key word) listing

Functioning test
Leak test
Sensitivity
Velocity

Custodian:
Army AR

Preparing Activity:
Army - AR

(Project 1315-AD21)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-71030(AR)	920624

3. DOCUMENT TITLE: CARTRIDGE, IGNITION, M981 LOADING, ASSEMBLING, AND PACKING

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: SMACA-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.